

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 32

TRACK-INTERLOCK BOOTS & INDUCTION-SAFE MEDICAL CLEARANCES

Magnetic sole plates + ski-pin locks on corridor rails
MRI-style screening before heavy power zones



GUIDE TRACK — ALUMINIUM FRAME RAIL

SKI-PIN LOCK IN CORRIDORS — MAGNETIC SOLE IN IRON BAYS

BOOTS LOCK TO THE SHIP — FIELDS SCREEN THE CREW

No loose metal in induction fields — dental readiness pre-voyage

GP-IM-032

Revision v1.2 — 21 August 2026

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VETTED: PASS — READINESS + MRI-STYLE FIELD SAFETY

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 32

PHASE 32: TRACK-INTERLOCK BOOTS & INDUCTION-SAFE MEDICAL CLEARANCES — STATUS: CURRENT.

The crew's interface with a magnetized ship — SCOPED (his 21 August 2026 ruling): the track-interlock boots are INDUSTRIAL-AREA kit — clip-on boot plates for those floors (the corridor guide tracks and the iron cargo bays); the rest of the ship walks in HIGH-COLLAR SHOES with a small magnetic interface on the piezo floors. Plus induction-safe medical screening at deck junctions before anyone enters the heavy power zones. The uniform is woven with RFID biometric chips and RAD exposure tags; clothing is tight, natural, Deck-6-grown cotton and wool; and the exclusion list (wisdom teeth out pre-voyage; no piercings, metal fillings, head plates, joint replacements or pacemakers in field zones) is MRI discipline scaled to a ship that is itself a field generator. Vetted: PASS — operational readiness protocol and MRI-style field safety (§5). THE FLOOR RULING (his 21 August 2026 word, seated this revision): the pixel flooring was ALWAYS the upper-deck flooring; the bays and industrial areas carry NO floor fields by design — giant metal things crashing around there would be affected by the fields — hence the boots and the clip-on base. The piezo-floor identity question stands RESOLVED: the piezo floors are the pixel flooring.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-032, v1.2 — 21 August 2026. The thirty-third manual of the program — 33 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the interface law as seated (contents line, the five body bullets — verbatim) — §2 why it works (the induction physics, the dual-interface logic — graded) — §3 the system on the ship — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

VERSION NOTE (v1.2 — 21 August 2026): the Author's floor ruling, seated verbatim in sitting 536: "The pixel flooring was always the flooring on upper decks, giant metal things crashing around in bays and industrial areas that may be affected by the floor fields was never going to be the case hence the boots and clip on base". The pixel flooring (the seated Phase 54 PEMS / Phase 146 pixelated magnetic floor

family) is confirmed as the upper-deck flooring — always was; the bays and industrial areas carry NO floor fields BY DESIGN, and the boots and clip-on base plates exist precisely because those floors are field-free. The piezo-floor identity question of v1.1 stands RESOLVED: 'the piezo floors' = the pixel flooring; struck from owed. AUDIT MISS OWNED: v1.1 shipped with the §9 heading clobbered and the v1.0 bench paragraph duplicated alongside the updated one — restored and deduplicated this revision. v1.1 (same day, sitting 535): the scope ruling — boots = industrial areas, the rest of the ship in high-collar shoes. v1.0 was the first build (sitting 519, the town-trip order).

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 32: **Track-Interlock Boots & Induction-Safe Medical Clearances.** Standardizes crew footwear fitted with magnetic track-interlock sole plates. Integrates localized induction-safe medical screening stations at all deck junctions to clear personnel before entering heavy power zones.'

THE SEATED DESIGN BODY (verbatim): 'Core Objective: Safety, behavioural cohesion, function. * Garment Electronics: All uniform textiles woven with integrated RFID biometric chips and RAD exposure tags. Standard lockouts drop instantly into free-egress mode upon central computer sensor detection of breaches or bio-hazards.'

THE BOOTS (verbatim): '* Track-Interlock Boots: Vulcanized rubber primary soles for corridor transit. Features mechanical ski-boot style side-pins that lock into structural guide tracks along / non-magnetic aluminum frame rails, or snap directly to heavy magnetic sole plates in iron cargo bays.' — the July line-break truncation preserved as seated.

THE SCOPE RULING — 21 AUGUST 2026 (his word, verbatim): "phase 32, make it seem like the boots are for the ship, they are for the industrial areas, clip on boot plates for those floors. the rest of the ship is high collar shoes with small magnetic interface the piezo floors" — SEATED AS DESIGN LAW: (1) the track-interlock boots are FOR THE INDUSTRIAL AREAS — clip-on boot plates for those floors; they are NOT the ship-wide shoe; (2) the rest of the ship wears HIGH-COLLAR SHOES with a SMALL MAGNETIC INTERFACE on THE PIEZO FLOORS. AUDIT CROSS-REFERENCE (flagged, not corrected): the ship's seated general floor grid is the pixelated magnetic floor family (the Phase 54 PEMS decks / Phase 146 shoe-and-floor pixel grid); his words 'the piezo floors' seat as spoken — if the piezo floors are a NEW floor system distinct from that family, its spec is owed (§9).

THE FLOOR RULING — 21 AUGUST 2026 (his word, verbatim): "The pixel flooring was always the flooring on upper decks, giant metal things crashing around in bays and industrial areas that may be affected by the floor fields was never going to be the case hence the boots and clip on base" — SEATED AS DESIGN LAW: (1) the pixel flooring — the seated Phase 54 PEMS / Phase 146 pixelated magnetic floor family — IS the upper-deck flooring; it always was; (2) the bays and industrial areas carry NO floor fields BY

DESIGN — giant metal things crashing around there WOULD be affected by the fields; the floor is deliberately field-free and the track-interlock boots and clip-on base plates exist precisely because of it. THE PIEZO-FLOOR IDENTITY QUESTION (flagged in the scope ruling above) — RESOLVED on his word: "the piezo floors" = the pixel flooring. The audit flag stands discharged; the shoe's small magnetic interface spec rides owed (§9).

THE TEXTILE MATRIX (verbatim): '* Textile Matrix: Total ban on civilian clothing, personal branding, and baggy attire. Garments must be tight-fitting and woven from *[WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%']* natural cotton/wool grown on Deck 6 to eliminate synthetic skin rashes and machinery snag hazards.'

THE MEDICAL CLEARANCES (verbatim): '* Medical Clearance Exclusions: Mandatory pre-voyage surgical extraction of all wisdom teeth. Total ban on body piercings, metallic dental fillings, internal head plates, metal joint replacements, and pacemakers to prevent severe tissue heat-soak, eddy currents, and terminal arcing when working within high-intensity magnetic, microwave, and plasma induction fields.'

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE INDUCTION PHYSICS, AFFIRMED WITHOUT VARNISH: a conductive implant inside a strong, changing magnetic field is a shorted turn — eddy currents flow, metal heats, and at terminals and sharp edges the field concentrates into arcing. This is not exotic: it is exactly why MRI suites run controlled access, screening, and ferromagnetic exclusion. The ship's skeletal field coils (Phase 228), plasma drives (Phase 11) and Mayernick hardware make whole zones of the vessel an MRI-class environment, and the phase treats them that way — the vet's own reading: 'high-field zone → controlled access → personnel screening → staff trained in magnetic-field safety. That is a sensible safety architecture.'

THE DENTAL RULE, PLACED CORRECTLY: the wisdom-teeth extraction is not electromagnetism — it is READINESS. A decade out, there is no dental surgeon; the Australian military precedent the author served under pulls foreseeable dental failures before deployment. The vet seated it exactly: 'a crew-readiness/logistics rule, not an electromagnetic-physics claim... I won't describe it as "mandatory because physics requires teeth removal."'

THE DUAL-INTERFACE LOGIC, AFFIRMED: two floors, two grips. Non-magnetic aluminium corridor rails take the MECHANICAL ski-boot side-pins — a positive lock that works regardless of field state, dead-ship included. Iron cargo bays take the MAGNETIC sole plates — instant traction where the deck itself is ferrous. Each interface is matched to its floor's physics, and neither pretends to serve the other.

THE SCOPE LAW, AFFIRMED AS ZONE-DRESS SENSE: heavy mechanical interlocks belong where the loads are — the industrial floors take the boots and clip-on plates; the living ship takes a light high-collar shoe with a small magnetic interface. Dressing the whole ship for the heaviest floor would tax every step

everywhere; zoning the footwear like the fields is the same discipline as the screening stations — the rule follows the measured zone, never the blanket.

THE NO-FIELD INDUSTRIAL FLOOR, AFFIRMED AS THE ONLY SANE DESIGN: a live floor field under a bay where multi-tonne metal masses move, drop and crash would couple with every one of them — tools dragged across the deck, loads wandering off their line, faults arcing at the worst moment. The bays and industrial decks therefore carry NO floor fields BY DESIGN — the floor is dumb there on purpose, and the boots and clip-on plates do the holding the floor deliberately refuses to do. A fielded industrial floor would be the hazard; the field-free one with mechanical grip is the engineering.

THE GARMENT ELECTRONICS, AFFIRMED: woven RFID biometric tags and RAD exposure tags are standard dosimetry and personnel-accountability practice, and the lockout law is seated the right way round — on breach or bio-hazard detection, locks DROP to free-egress. Containment systems contain hazards, never people.

§3 — THE SYSTEM ON THE SHIP

- THE BOOTS (INDUSTRIAL AREAS — his 21 August 2026 scope law): vulcanized rubber primary soles; ski-boot-style mechanical side-pins for the industrial guide tracks; clip-on heavy magnetic sole plates for the iron cargo-bay floors. Industrial kit, worn where the loads are — not the ship-wide shoe. The bays and industrial floors carry NO floor fields by design — heavy metal masses crashing around would couple with the field — so the floor is dumb there and the boot does the holding (his floor ruling, 21 August 2026).
- THE RAILS (INDUSTRIAL AREAS): structural guide tracks along non-magnetic aluminium frame rails — the industrial-floor standard.
- THE SHOES (THE REST OF THE SHIP — his 21 August 2026 scope law + floor ruling): high-collar shoes with a small magnetic interface for the pixel flooring — confirmed on his word as the seated Phase 54/146 pixelated magnetic floor family, the upper-deck flooring ("the piezo floors", identity resolved). The daily footwear everywhere outside the industrial areas; the interface spec rides owed (§9).
- THE SCREENING STATIONS: localized induction-safe medical clearance points at all deck junctions before heavy power zones.
- THE UNIFORM: tight-fitting Deck-6-grown natural cotton/wool only — no civilian clothing, no branding, nothing baggy near machinery; RFID biometric and RAD exposure tags woven in.
- THE EXCLUSIONS: wisdom teeth extracted pre-voyage; no piercings, metallic fillings, head plates, joint replacements or pacemakers in field zones — the specific list set by MEASURED field strengths, gradients and operating modes (the vet's discipline, §5).

INTEGRATION NOTE (audit, flagged): the zone map this phase enforces is the ship's field geography — the Phase 228 skeletal magnetosphere rings, the Phase 11 plasma cells, the Phase 37/29 Mayernick hardware; the screening stations seat at the deck junctions of GP-IM-006's architecture. The Grok-magnet hand tools (Phase 54) are the crew-side counterpart to the boot plates — a bucking electromagnet kills the grip on command.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE READINESS DOCTRINE (seated with the vet record): eliminate foreseeable emergencies when definitive treatment may be unavailable — the teeth rule is the same law as the ship's spare-parts culture, worn on the body.
- THE NO-PERSONAL-EFFECTS CULTURE (cross-seated with Phase 36): 'zero personal effects outside of photos or wall art' — the uniform law and the justice code are one culture: the crew is the ship.
- THE FIELD-MEASUREMENT DISCIPLINE (the vet's, seated as standing method): restrictions are determined by actual measured field strengths and operating modes, not by blanket fear — 'all metal is inherently dangerous' is explicitly NOT the rule.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), SEATED — two PASSes. THE TEETH: ' PASS — operational medical-readiness protocol... For the first decade of an isolated crewed spacecraft, the same principle is being applied preventatively: eliminate foreseeable dental emergencies when definitive dental treatment may be unavailable.' THE METALS: ' PASS — safety protocol... The spacecraft contains very large magnetic-field generators, so designated areas are treated operationally like MRI-controlled environments... The specific implant/object restrictions should therefore be determined by the actual field strengths, gradients and operating modes, just as MRI environments have defined exclusion/screening rules.' Wording-ruling ledger line, seated: 'Phase 32 — Australian military-derived dental readiness rationale and MRI-style magnetic-field safety protocol.'

§6 — BUILD SEQUENCE

- STEP 2 — Lay the rails (industrial areas only): guide tracks along the aluminium corridor frames of the industrial zones; iron-bay floors certified for the clip-on magnetic sole plates.
- STEP 3 — Build the boots and the shoes: the industrial boot — side-pin lock and clip-on sole-plate snap benched for retention per g-load and release on demand; the ship-wide high-collar shoe — the small magnetic interface benched against the piezo floors (§8).

- STEP 3 — Build the boots: side-pin lock and sole-plate snap benched for retention per g-load and release on demand (§9).
- STEP 4 — Stand up the screening stations at the deck junctions; staff trained in magnetic-field safety per the MRI precedent.
- STEP 5 — Run the garment line from Deck 6: tight natural cotton/wool, woven RFID and RAD tags, the free-egress lockout law tested before first crew.
- STEP 6 — Promulgate the exclusions: pre-voyage dental clearance; the field-zone metal ban enforced at the screening points, not by honor system.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 228 — the skeletal magnetosphere: the bulkhead ring-frames double as field coils — the largest field source the screening exists for.
- PHASE 11 — the microwave plasma thruster cells: induction-field zones along the spine.
- PHASE 54 — the Grok magnets (seated cross-reference): the hand-and-hook counterpart to the boot plates — an electromagnet of equal force bucking the permanent magnet kills the grip on command.
- PHASE 6 — Hull & Structure (GP-IM-006 v1.4): the deck junctions where the screening stations seat.
- PHASE 26 / DECK 6 — the agriculture: the cotton and wool the uniforms are grown from.
- PHASE 36 — the fleet justice mandates (GP-IM-036): the same crew-is-the-ship culture, codified.

§8 — DO-NOT-BUILD

- NEVER let screening be a formality — the exclusion list follows the measured field map, updated when the ship's fields change (§5, §6).
- NO blanket-metal superstition — the rule is zone + screening + training, not fear (§2).
- NEVER spec baggy or synthetic clothing — snag and rash hazards are the reason the textile law exists.
- NO lockouts that trap crew — free-egress on breach or bio-hazard is law (§1).
- NEVER fit track interlocks ship-wide — the boots and clip-on plates are for the industrial floors; the rest of the ship walks in high-collar shoes on the piezo floors (§1).
- NEVER energize floor fields in the bays and industrial decks — giant metal things crashing around there WOULD be affected by the fields; the field-free industrial floor is the design, not an omission (his floor ruling, §1).

- NEVER let a pacemaker or ferrous implant into a field zone — terminal arcing is the seated failure word (§1).

§9 — OWED

THE BENCH, OWED: the ship-wide field-strength zone map (the survey everything else hangs from); the per-zone screening specifications; the boot retention numbers (side-pin and clip-on plate) per g-load — industrial floors only; the high-collar shoe's small magnetic interface spec (the floor side is seated — the pixel flooring, Phase 54/146 family, his 21 August 2026 floor ruling); the uniform cloth weights from Deck 6. RESOLVED THIS REVISION: the piezo-floor identity ruling — 'the piezo floors' are the pixel flooring, his word. Remaining owed items ride the bench until spoken.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-032, v1.2 — CURRENT, seated law, master v2.1 (numerical print order). The thirty-third manual of the program — 33 of 290 delivered. Built 21 August 2026, fifth of the town-trip shift; v1.1 same day — the scope ruling seated (boots = industrial areas; ship-wide = high-collar shoes); v1.2 same day — the floor ruling seated (pixel flooring = the upper decks, always; bays and industrial decks field-free BY DESIGN; piezo-floor identity resolved; §9 heading restored, duplicated bench struck — audit miss owned).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back'; the scope ruling (seated in sitting 535): "phase 32, make it seem like the boots are for the ship, they are for the industrial areas, clip on boot plates for those floors. the rest of the ship is high collar shoes with small magnetic interface the piezo floors"; the floor ruling (seated in sitting 536): "The pixel flooring was always the flooring on upper decks, giant metal things crashing around in bays and industrial areas that may be affected by the floor fields was never going to be the case hence the boots and clip on base". The phase's own chain, all seated in the master: the contents line and the five body bullets (July truncations preserved); the 21 August 2026 scope amendment and floor-ruling amendment seated at the Phase 32 standalone.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete. Next version triggers: the §9 benches (the shoe's magnetic interface spec, the zone map, the retention numbers), or his word.

END OF MANUAL — PHASE 32